COHOMOLOGICAL DIMENSION OF AN AFFINE SCHEME OF FINITE TYPE OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including both interludes and the final diagram; the Xy-pic diagram is delivered as the author wrote it.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Cohomological dimension cd(X) of a scheme and the vanishing of torsion-sheaf cohomology. Same chapter, the definition of the cohomological dimension as the supremum of the degrees in which torsion sheaves can have nonzero cohomology. Supplies the quantity bounded in the statement and the reduction of the proof to vanishing statements for torsion sheaves.
- Vanishing of higher direct images along a finite morphism. Same chapter, the result that a finite morphism has vanishing higher direct images. Supplies the case $\dim(X) = 0$, where the structure morphism is finite.
- Noether normalization for affine schemes of finite type over a field. Varieties, the Noether normalization producing a finite morphism from an affine variety to an affine space. Supplies the reduction of the general case to a smooth situation of the same dimension.
- Vanishing for curves and the curve case of the bound. Same chapter, the bound cd(X) at most 1 + cd(K) for a one-dimensional affine scheme of finite type over K. Supplies the base of the induction on the dimension.

Source reference. Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Navigation pointer invoked at proof line

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19775; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0732: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19778–19778. Navigation pointer invoked at proof line 19778; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0S: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19651–19651. Navigation pointer invoked at proof line 19782; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0CBI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19788–19788. Navigation pointer invoked at proof line 19788; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Navigation pointer invoked at proof line 19804; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19692–19692. Navigation pointer invoked at proof line 19815; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0T: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19673–19673. Navigation pointer invoked at proof line 19826; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0R: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19633–19633. Navigation pointer invoked at proof line 19827; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A21: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19835–19835. Navigation pointer invoked at proof line 19835; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F0N: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 13091–13091. Navigation pointer invoked at proof line 19842; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 04CA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6702–6702. Navigation pointer invoked at proof line 19845; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Navigation pointer invoked at proof line 19850; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 095U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18786–18786. Navigation pointer invoked at proof line 19873; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let K be a field. Let X be an affine scheme of finite type over K. Then we have $cd(X) \leq \dim(X) + cd(K)$.*

Proof. We will prove this by induction on $\dim(X)$. Let $\mathcal{F}$ be an abelian torsion sheaf on $X_{étale}$.

The case $\dim(X) = 0$. In this case the structure morphism $f : X \to \mathrm{Spec}(K)$ is finite. Hence we see that $R^i f_* \mathcal{F} = 0$ for $i > 0$, see Proposition 03QP. Thus $H^i_{étale}(X, \mathcal{F}) = H^i_{étale}(\mathrm{Spec}(K), f_*\mathcal{F})$ by the Leray spectral sequence for f (Cohomology on Sites, Lemma 0732) and the result is clear.

The case $\dim(X) = 1$. This is Lemma 0F0S.

Assume $d = \dim(X) > 1$ and the proposition holds for finite type affine schemes of dimension $< d$ over fields. By Noether normalization, see for example Varieties, Lemma 0CBI, there exists a finite morphism $f : X \to \mathbf{A}^d_K$. Recall that $R^i f_* \mathcal{F} = 0$ for $i > 0$ by Proposition 03QP. By the Leray spectral sequence for f (Cohomology on Sites, Lemma 0732) we conclude that it suffices to prove the result for $\pi_* \mathcal{F}$ on $\mathbf{A}^d_K$.

Interlude I. Let $j : X \to Y$ be an open immersion of smooth d-dimensional varieties over K (not necessarily affine) whose complement is the support of an effective Cartier divisor D. The sheaves $R^q j_* \mathcal{F}$ for $q > 0$ are supported on D. We claim that $(R^q j_* \mathcal{F})_{\overline{y}} = 0$ for $a = \mathrm{trdeg}_K(\kappa(y)) > d - q$. Namely, by Theorem 03Q9 we have

$$(R^q j_* \mathcal{F})_{\overline{y}} = H^q(\mathrm{Spec}(\mathcal{O}^{sh}_{Y,y}) \times_Y X, \mathcal{F})$$

Choose a local equation $f \in \mathfrak{m}_y \subset \mathcal{O}_{Y,y}$ for D. Then we have

$$\mathrm{Spec}(\mathcal{O}^{sh}_{Y,y}) \times_Y X = \mathrm{Spec}(\mathcal{O}^{sh}_{Y,y}[1/f])$$

Using Lemma 0F0U we get an embedding

$$K(t_1, \ldots, t_a)^{sep}(x) = K(t_1, \ldots, t_a)^{sep}[x]_{(x)}[1/x] \longrightarrow \mathcal{O}^{sh}_{Y,y}[1/f]$$

Since the transcendence degree over K of the fraction field of $\mathcal{O}^{sh}_{Y,y}$ is d, we see that $\mathcal{O}^{sh}_{Y,y}[1/f]$ is a filtered colimit of $(d - a - 1)$-dimensional finite type algebras over the field $K(t_1, \ldots, t_a)^{sep}(x)$ which itself has cohomological dimension 1 by Lemma 0F0T. Thus by induction hypothesis and Lemma 0F0R we obtain the desired vanishing.

Interlude II. Let Z be a smooth variety over K of dimension $d - 1$. Let $E_a \subset Z$ be the set of points $z \in Z$ with $\mathrm{trdeg}_K(\kappa(z)) \leq a$. Observe that E_a is closed under

specialization, see Varieties, Lemma 0A21. Suppose that $\mathcal{G}$ is a torsion abelian sheaf on Z whose support is contained in E_a. Then we claim that $H^b_{étale}(Z, \mathcal{G}) = 0$ for $b > a + \mathrm{cd}(K)$. Namely, we can write $\mathcal{G} = \mathrm{colim}\, \mathcal{G}_i$ with $\mathcal{G}_i$ a torsion abelian sheaf supported on a closed subscheme Z_i contained in E_a, see Lemma 0F0N. Then the induction hypothesis kicks in to imply the desired vanishing for $\mathcal{G}_i$[1]. Finally, we conclude by Theorem 09YQ.

Consider the commutative diagram

$$\begin{array}{ccc} \mathbf{A}^d_K & \xrightarrow{j} & \mathbf{P}^1_K \times_K \mathbf{A}^{d-1}_K \\ & {\scriptstyle f} \searrow \quad \swarrow {\scriptstyle g} & \\ & \mathbf{A}^{d-1}_K & \end{array}$$

Observe that j is an open immersion of smooth d-dimensional varieties whose complement is an effective Cartier divisor D. Thus we may use the results obtained in interlude I. We are going to study the relative Leray spectral sequence

$$E_2^{p,q} = R^p g_* R^q j_* \mathcal{F} \Rightarrow R^{p+q} f_* \mathcal{F}$$

Since $R^q j_* \mathcal{F}$ for $q > 0$ is supported on D and since $g|_D : D \to \mathbf{A}^{d-1}_K$ is an isomorphism, we find $R^p g_* R^q j_* \mathcal{F} = 0$ for $p > 0$ and $q > 0$. Moreover, we have $R^q j_* \mathcal{F} = 0$ for $q > d$. On the other hand, g is a proper morphism of relative dimension 1. Hence by Lemma 095U we see that $R^p g_* j_* \mathcal{F} = 0$ for $p > 2$. Thus the E_2-page of the spectral sequence looks like this

$$\begin{array}{ccc} g_* R^d j_* \mathcal{F} & 0 & 0 \\ \ldots & \ldots & \ldots \\ g_* R^2 j_* \mathcal{F} & 0 & 0 \\ g_* R^1 j_* \mathcal{F} & 0 & 0 \\ g_* j_* \mathcal{F} & R^1 g_* j_* \mathcal{F} & R^2 g_* j_* \mathcal{F} \end{array}$$

We conclude that $R^q f_* \mathcal{F} = g_* R^q j_* \mathcal{F}$ for $q > 2$. By interlude I we see that the support of $R^q f_* \mathcal{F}$ for $q > 2$ is contained in the set of points of $\mathbf{A}^{d-1}_K$ whose residue field has transcendence degree $\leq d - q$. By interlude II

$$H^p(\mathbf{A}^{d-1}_K, R^q f_* \mathcal{F}) = 0 \text{ for } p > d - q + \mathrm{cd}(K) \text{ and } q > 2$$

On the other hand, by Theorem 03Q9 we have $R^2 f_* \mathcal{F}_{\overline{\eta}} = H^2(\mathbf{A}^1_{\overline{\eta}}, \mathcal{F}) = 0$ (vanishing by the case of dimension 1) where η is the generic point of $\mathbf{A}^{d-1}_K$. Hence by interlude II again we see

$$H^p(\mathbf{A}^{d-1}_K, R^2 f_* \mathcal{F}) = 0 \text{ for } p > d - 2 + \mathrm{cd}(K)$$

Finally, we have

$$H^p(\mathbf{A}^{d-1}_K, R^q f_* \mathcal{F}) = 0 \text{ for } p > d - 1 + \mathrm{cd}(K) \text{ and } q = 0, 1$$

by induction hypothesis. Combining everything we just said with the Leray spectral sequence $H^p(\mathbf{A}^{d-1}_K, R^q f_* \mathcal{F}) \Rightarrow H^{p+q}(\mathbf{A}^d_K, \mathcal{F})$ we conclude. □

[1]Here we first use Proposition 04CA to write $\mathcal{G}_i$ as the pushforward of a sheaf on Z_i, the induction hypothesis gives the vanishing for this sheaf on Z_i, and the Leray spectral sequence for $Z_i \to Z$ gives the vanishing for $\mathcal{G}_i$.